



Bongakonke Mngadi

Johannesburg, Bramley, forest 96

+27655829095 | mngadibongakonke@gmail.com

🌐 <https://bongakonke-mngadi-portfolio.vercel.app/>

in <https://www.linkedin.com/in/bonga-ngema-a56834320/>

🐦 https://x.com/bonga_ngema_0

Objective

AI Agent Engineer and Full-Stack Developer with strong experience building autonomous, production-grade AI systems that do more than answer questions, they take actions, complete workflows, integrate with real business tools, and deliver measurable operational value. I specialize in designing and deploying AI agents that use LLM reasoning, tool calling, browser automation, workflow orchestration, APIs, messaging platforms, databases, and human-in-the-loop approval systems to automate real-world business processes. My work spans autonomous browser agents, WhatsApp sales agents, AI receptionists, AI document processors, recruiter follow-up agents, multilingual video explanation systems, negotiation agents, educational tutors, and growth automation systems. Currently building AI agents for business clients at BBD Johannesburg, including client-facing systems for medical practices, car dealerships, accounting firms, and real estate sales teams. Previously worked as an AI Engineer / AI Agent Engineer Graduate Intern at MTN, contributing to AI automation, intelligent workflow systems, and early-stage enterprise AI experimentation. Highly skilled in Python, JavaScript, TypeScript, React, Node.js, Express, Firebase, Supabase, OpenAI APIs, LLM orchestration, agentic workflows, browser automation, WhatsApp/Twilio integrations, Google APIs, CRM automation, vector search, RAG, and cloud deployment. Also highly proficient with modern AI engineering tools including Cursor, Claude Code, OpenAI Codex, GitHub Copilot, Windsurf, ChatGPT, Claude, and agent-assisted development workflows.

Experience

- MTN — Graduate Internship 2022 / 2023** 2022 - 2023
AI Engineer Graduate Intern
Worked as part of MTN's graduate internship programme with a focus on AI engineering, automation, and intelligent systems. Contributed to internal AI experiments and automation workflows designed to improve productivity, data handling, and intelligent customer/business operations.

Key Responsibilities:
 - Supported the development and experimentation of AI-powered tools and workflow automation systems.
 - Built and tested early AI agent prototypes capable of understanding user intent and triggering actions.
 - Assisted with backend development, API integrations, and data-driven automation workflows.
 - Worked with Python and JavaScript-based systems for automation, data processing, and AI workflow prototyping.
 - Explored LLM-based solutions for enterprise use cases including internal assistants, document automation, customer-support workflows, and operational intelligence.
 - Collaborated with technical teams to understand business requirements and translate them into functional AI prototypes.
 - Assisted in testing, debugging, and improving reliability of AI-driven workflows.
 - Gained experience in enterprise software practices, stakeholder communication, and built practical experience applying AI to real business problems inside a large telecommunications environment.
 - Developed strong foundations in API-based AI systems, automation workflows, and production software delivery.
 - Improved ability to communicate technical AI concepts to business and engineering stakeholders.
 - Strengthened skills in Python, JavaScript, backend APIs, cloud tools, and AI-assisted automation, documentation, and production-readiness thinking.
- BBD — Johannesburg** February-2024 - Present
AI Agent Engineer / Full-Stack Developer
Building client-facing AI agent systems for South African businesses, focused on automating operational workflows that normally require human admin, sales, receptionist, or back-office support. Work involves designing AI systems that can communicate with users, understand business context, call tools, update systems, schedule events, process documents, and escalate to humans when needed.

Key Responsibilities:

- Design and build full-stack AI agent applications for business clients using React, Node.js, Express, Firebase, Supabase, Python, OpenAI-compatible APIs, Twilio, Google APIs, and CRM integrations.
- Build agents that perform real actions instead of simply generating text responses, including booking appointments, qualifying leads, updating databases, sending reminders, processing documents, and logging customer interactions.
- Develop WhatsApp-first AI workflows for businesses where customers naturally communicate through messaging.
- Create reusable AI agent architecture patterns including:
 - intake and qualification flows
 - tool-calling layers
 - calendar booking flows
 - follow-up engines
 - CRM logging
 - document extraction
 - human escalation
 - audit logging
 - permission-based approval flows
- Integrate LLMs with external APIs such as Google Calendar, Gmail, Google Sheets, Supabase, Firebase, Twilio, and CRM-style systems.
- Build dashboards that allow businesses to monitor leads, appointments, conversations, agent actions, escalations, and workflow status.
- Use modern AI developer tools such as Cursor, Claude Code, OpenAI Codex, GitHub Copilot, ChatGPT, Claude, and Windsurf to accelerate development, debugging, code review, and architecture design.
- Work across the full product lifecycle: requirements analysis, system architecture, frontend development, backend APIs, database schema design, AI workflow logic, testing, deployment, and iteration.

Selected Client AI Agent Systems at BBD:

Real Estate AI Sales Agent:

Built an AI sales agent for real estate workflows that automates lead qualification, property matching, viewing bookings, follow-ups, reminders, and CRM logging.

Business problem solved:

Real estate agents lose hours every day replying to repetitive enquiries, qualifying buyers, checking calendars, sending viewing confirmations, and following up manually.

What the agent does:

- Receives buyer enquiries through WhatsApp or web channels.
- Qualifies the lead based on budget, area, bedrooms, buying intent, and timeline.
- Recommends suitable property listings from available inventory.
- Checks viewing availability through calendar integrations.
- Books property viewings automatically.
- Sends confirmation messages and reminders.
- Follows up after viewings.
- Escalates hot buyers to a human agent with a structured lead summary.
- Logs conversations, property interest, bookings, and follow-up status into a dashboard or CRM.

Technologies used:

- React
- Node.js
- Express
- Python
- Twilio WhatsApp API
- Supabase
- Firebase Auth
- Google Calendar API
- Google Sheets API
- HubSpot-style CRM workflows
- OpenAI-compatible LLM APIs

- Webhooks
- Docker
- Vercel

Education

- **University of Cape Town** 2021
Bachelor degree in Computer science
88
- **University of Cape Town** 2022
Honours degree in Computer Science
72.5

Skills

- Programming Languages(Python - JavaScript - TypeScript - SQL - HTML - CSS)
- AI / LLM Engineering(OpenAI API - Claude / Anthropic workflows - GPT reasoning models - Kimi K2.6-style OpenAI-compatible APIs - Prompt engineering - Function calling - Tool use - Agent orchestration - Multi-step reasoning flows - RAG pipelines - Vector embeddings - Structured output parsing - Human-in-the-loop approval systems - Agent memory systems - Vision-enabled AI workflows - Document intelligence - AI workflow evaluation and debugging)
- AI Coding / Developer Tools(Cursor - Claude Code - OpenAI Codex - GitHub Copilot - Windsurf - ChatGPT - Claude - VS Code - Postman - Git - GitHub - Vercel - Netlify)
- Frontend(React - Vite - Tailwind CSS - JavaScript ES6+ - TypeScript - Responsive UI development - Component-based architecture - Dashboard development - Landing page development - Authentication flows - UX-focused product design)
- Backend(Node.js - Express.js - REST APIs - WebSockets - Firebase Admin - Supabase - PostgreSQL - Firestore - Authentication middleware - API integrations - Background workflows - Cron-style automation - Error handling and observability)
- Automation / Agent Tools(Playwright - Browser automation - Twilio WhatsApp API - Meta WhatsApp Cloud API - Google Calendar API - Gmail API - Google Sheets API - HubSpot-style CRM workflows - Stripe API - SendGrid - OCR / vision extraction - FFmpeg - Webhooks - Event-driven workflows)
- Databases / Storage(- Supabase - PostgreSQL - Firebase Firestore - Google Sheets - IndexedDB - Vector databases - In-memory demo stores - File-based fallback systems)
- Cloud / Deployment(Vercel - Netlify - Railway-style deployment workflows - Render-style deployment workflows - Docker - GitHub repositories - Environment variables - Production build pipelines - CORS/API configuration)

Projects

- **Handio-General AI Browser Agent**
Live: <https://handio-lab.vercel.app/>

Autonomous AI browser agent that opens real websites, reads pages, reasons through tasks, and completes actions inside a browser like a human operator.

Highlights:

- Browser automation using Playwright-style workflows.
- AI reasoning loop for planning, acting, observing, and verifying.
- Human-in-the-loop flows for sensitive actions.
- WebSocket-based runtime updates.
- Designed for real-world task automation across websites.

Tech Stack:

- React
- Vite
- Node.js
- TypeScript
- Playwright
- WebSockets
- Redis
- Docker

- OpenAI GPT-style reasoning
- Kimi-style multimodal reasoning

- **EstateFlow-AI Property Sales Agent**

Live: <https://ai-real-estate-agent-virid.vercel.app/>

WhatsApp-native real estate sales agent that turns property enquiries into booked viewings automatically.

Highlights:

- Qualifies buyers through WhatsApp.
- Recommends matching properties.
- Books viewings using calendar integration.
- Sends reminders and follow-ups.
- Escalates hot leads with clean summaries.
- Includes dashboard for lead tracking.

Tech Stack:

- React
- Vite
- Node.js
- Express
- Python
- Twilio WhatsApp
- Firebase Admin
- Supabase
- Google Calendar API
- Google Sheets API
- HubSpot-style CRM workflows
- Docker

- **Annotated AI-Source-Linked Social Annotation Platform**

Live: <https://annotated-v2.vercel.app/>

A source-linked annotation platform where users can clip web text or YouTube timestamps and publish annotations with traceable source context.

Highlights:

- Chrome extension workflow.
- Public feed for annotations.
- Source-linked posts.
- Text clipping and YouTube timestamp clipping.
- AI-generated title, summary, tags, and fair-use note.
- Firebase-ready full-stack architecture.

Tech Stack:

- React
- Vite
- Chrome Extension MV3
- Node.js
- Express
- Firebase Auth
- Firestore
- IndexedDB
- WebRTC-style browser workflows
- Kimi/OpenAI-compatible LLM APIs

- **GrowthGhost-Autonomous Growth Marketer**

Live: <https://vukabrowser.vercel.app/>

Autonomous growth agent that connects website, analytics, and social channels, then plans and executes

growth experiments.

Highlights:

- Growth autopilot concept.
- Ghost Feed for visible agent activity.
- Analytics-informed decision-making.
- Content and experiment workflow automation.
- Waitlist and launch flow support.

Tech Stack:

- React
- Vite
- Firebase
- Firestore
- Google Analytics-style workflows
- LinkedIn/X/Reddit-style API concepts
- SendGrid
- Supabase
- WebSockets
- Node.js
- Docker

- **Handio v2-Autonomous Internet Worker**

Live: <https://sinterbrain-main-folder-pkyr.vercel.app/>

AI-powered explainer that generates personalized video explanations in the user's preferred language.

Highlights:

- Multilingual explanation generation.
- Personalized learning content.
- Visual explanation workflow.
- Language-first UX.
- Video explanation product concept.

Tech Stack:

- React
- Vite
- OpenAI GPT-style reasoning
- Nano Banana Pro Image Gen API
- Veo 3.1 Video Generation API
- ElevenLabs TTS API
- FFmpeg
- WebRTC
- Cloudflare Workers
- Redis
- Node.js
- TypeScript
- Docker

- **DealSmith-AI Negotiation Agent**

Live: <https://sinterbrain-main-folder-9rbk.vercel.app/>

AI agent that negotiates bills, rent, salary, bank fees, medical bills, service contracts, and other deal-based conversations.

Highlights:

- Goal-based negotiation.
- Walk-away price logic.
- Calm, professional negotiation strategy.

- Human approval before acceptance.
- Designed for consumer savings and business negotiation workflows.

Tech Stack:

- React
- Vite
- OpenAI GPT-style reasoning
- RLHF negotiation workflows
- Sentiment analysis
- NLP intent classification
- Twilio Voice-style integrations
- Stripe Billing API
- Redis
- BullMQ
- Node.js
- TypeScript
- Kubernetes
- Terraform

- **SA Science Tutor- AI Grade 12 Physical Science Tutor**

Live: ``https://physical-tutor.vercel.app/``

AI-powered tutor for South African Grade 12 Physical Science learners.

Highlights:

- CAPS-aligned explanations.
- Topic filtering.
- Step-by-step science tutoring.
- Learner-friendly explanations.
- Useful for Newton's Laws, electricity, chemistry, waves, organic chemistry, and exam prep.

Tech Stack:

- React
- Vite
- Python
- OpenAI GPT-style reasoning
- RAG pipeline
- CAPS curriculum embeddings
- LangChain
- Pinecone-style vector database
- LaTeX rendering
- Wolfram Alpha API
- Supabase
- Edge Functions
- Node.js
- TypeScript
- Vercel

Achievements & Awards

- - Built and deployed a portfolio of 8 production-style AI/full-stack products with live demos and polished user interfaces. - Built AI agent systems that automate real business operations across real estate, healthcare, automotive sales, accounting, education, recruitment, growth, and negotiation. - Developed WhatsApp-first AI agents capable of qualifying leads, booking appointments, sending reminders, and updating business systems. - Created autonomous browser-agent workflows capable of navigating websites and completing tasks. - Designed AI systems with human-in-the-loop approval for sensitive or irreversible actions. - Built full-stack dashboards for monitoring conversations, leads, bookings, agent actions, and workflow status. - Developed strong modern AI engineering workflow using Cursor, Claude Code, OpenAI Codex, GitHub Copilot, ChatGPT, Claude, and Windsurf - Built and deployed applications using Vercel, Netlify, Firebase, Supabase, Node.js, React, Python, and Twilio. - Created strong AI product demos suitable for startup, enterprise, and client-facing environments.

Interests

- Professional Value Proposition: I build AI agents that work like virtual employees. Not basic chatbots. Not FAQ bots. Not scripts that only reply. I build systems that: - understand user intent - collect missing information - call business tools - book appointments - update CRMs - send emails and WhatsApp messages - process documents - trigger follow-ups - escalate complex cases - maintain audit logs - complete tasks end-to-end My strength is combining AI reasoning, full-stack engineering, workflow automation, and product design into practical systems businesses can actually use.

Personal Details

- Gender : Male
- GitHub : <https://github.com/BongakonkeMngadi>

Additional Information

AI Agent Engineer and Full-Stack Developer with experience building production-grade AI systems that automate real business workflows. Skilled in Python, JavaScript, TypeScript, React, Node.js, Express, Firebase, Supabase, OpenAI APIs, Claude-style workflows, Twilio WhatsApp, Google APIs, browser automation, RAG, document intelligence, and modern AI coding tools including Cursor, Claude Code, OpenAI Codex, GitHub Copilot, and Windsurf. Built AI agents for real estate, medical practices, car dealerships, accounting firms, recruitment, education, growth automation, negotiation, and autonomous browsing. Strong ability to take an idea from business problem to working deployed product.